

CS4780 - Lab 2: Descriptive Statistics and Plotting

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Submitting your Lab assignment

Add the answers for the individual exercises to `Lab02_Worksheet.txt` available on BlackBoard. At the end of the lab, please submit the filled worksheet as **plain text** via the corresponding link on Blackboard. *No other formats will be considered.*

After submission deadline, the answers to the exercises will be published as `Lab02_Solutions.txt` on Blackboard.

Introduction

In this tutorial, we will look at descriptive statistics and different ways to visualise the data.

Getting the data

We will use the ‘mtcars’ dataset available in R:

```
my_data <-mtcars
```

The quickest way to get an overview is the `summary([...])` function:

```
summary(my_data)
```

```
##      mpg          cyl          disp          hp
## Min.   :10.40   Min.    :4.000   Min.    : 71.1   Min.    : 52.0
## 1st Qu.:15.43   1st Qu.:4.000   1st Qu.:120.8   1st Qu.: 96.5
## Median :19.20   Median :6.000   Median :196.3   Median :123.0
## Mean   :20.09   Mean    :6.188   Mean    :230.7   Mean    :146.7
```

```
## 3rd Qu.:22.80 3rd Qu.:8.000 3rd Qu.:326.0 3rd Qu.:180.0
## Max. :33.90 Max. :8.000 Max. :472.0 Max. :335.0
##      drat      wt      qsec      vs
## Min. :2.760 Min. :1.513 Min. :14.50 Min. :0.0000
## 1st Qu.:3.080 1st Qu.:2.581 1st Qu.:16.89 1st Qu.:0.0000
## Median :3.695 Median :3.325 Median :17.71 Median :0.0000
## Mean :3.597 Mean :3.217 Mean :17.85 Mean :0.4375
## 3rd Qu.:3.920 3rd Qu.:3.610 3rd Qu.:18.90 3rd Qu.:1.0000
## Max. :4.930 Max. :5.424 Max. :22.90 Max. :1.0000
##      am      gear      carb
## Min. :0.0000 Min. :3.000 Min. :1.000
## 1st Qu.:0.0000 1st Qu.:3.000 1st Qu.:2.000
## Median :0.0000 Median :4.000 Median :2.000
## Mean :0.4062 Mean :3.688 Mean :2.812
## 3rd Qu.:1.0000 3rd Qu.:4.000 3rd Qu.:4.000
## Max. :1.0000 Max. :5.000 Max. :8.000
```

We can also calculate descriptive statistics of variables such as the mean and standard deviation individually:

```
# mean
mean(my_data$mpg)
```

```
## [1] 20.09062
```

```
# standard deviation
sd(my_data$mpg)
```

```
## [1] 6.026948
```

Exercise 1

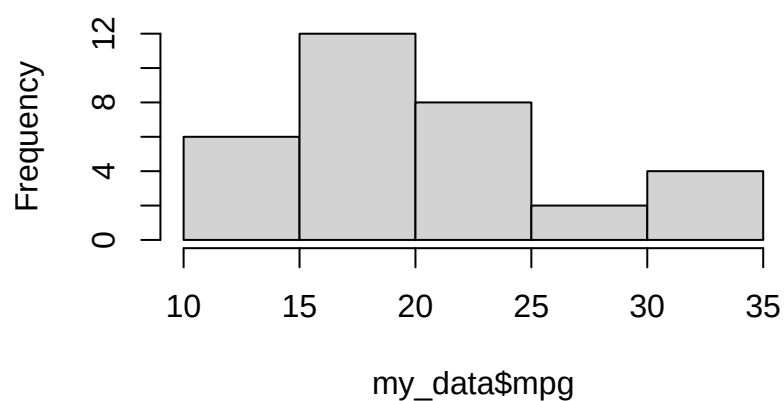
Using the `head` function, list the first 7 lines of the `my_data` dataframe. Tip: Get the help for the `head` function with `?head`.

Histograms

The most basic plot for a single variable is the frequency distribution histogram.

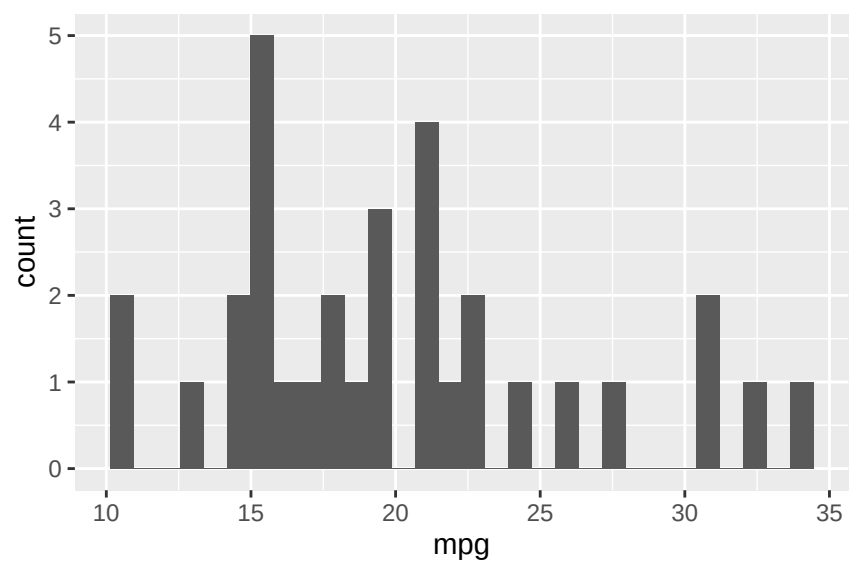
```
hist(my_data$mpg)
```

Histogram of my_data\$mpg

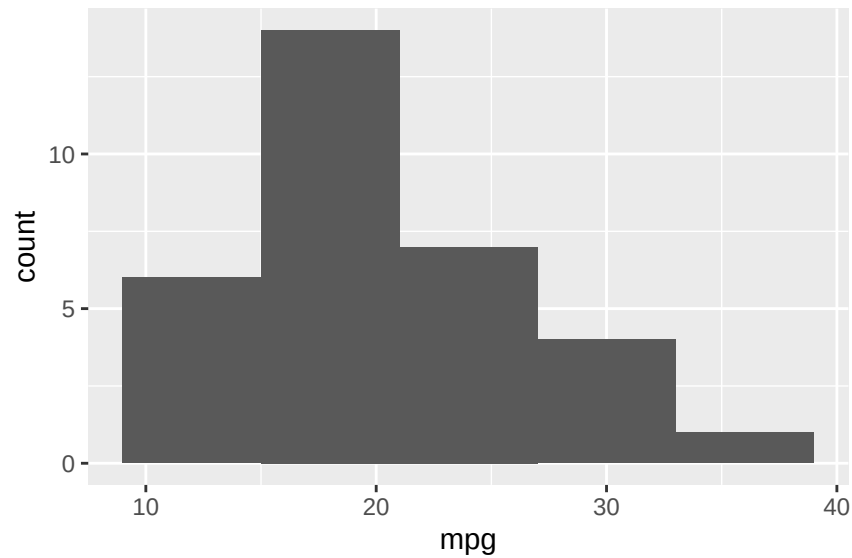


Here's an example to do the histogram using the 'ggplot' package, which gives you more control over the parameters and appearance:

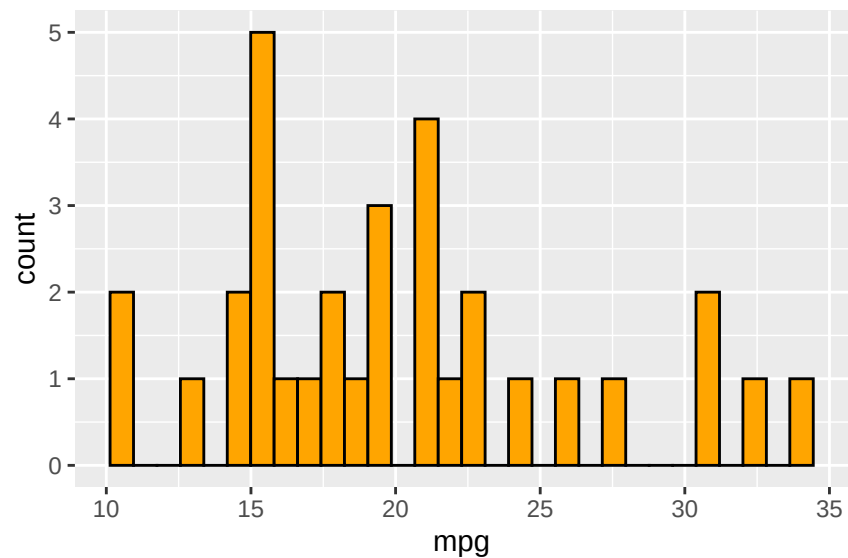
```
library(ggplot2)
# Basic histogram
ggplot(my_data, aes(x=mpg)) + geom_histogram()
```



```
# Change the width of bins
ggplot(my_data, aes(x=mpg)) +
  geom_histogram(binwidth=6)
```

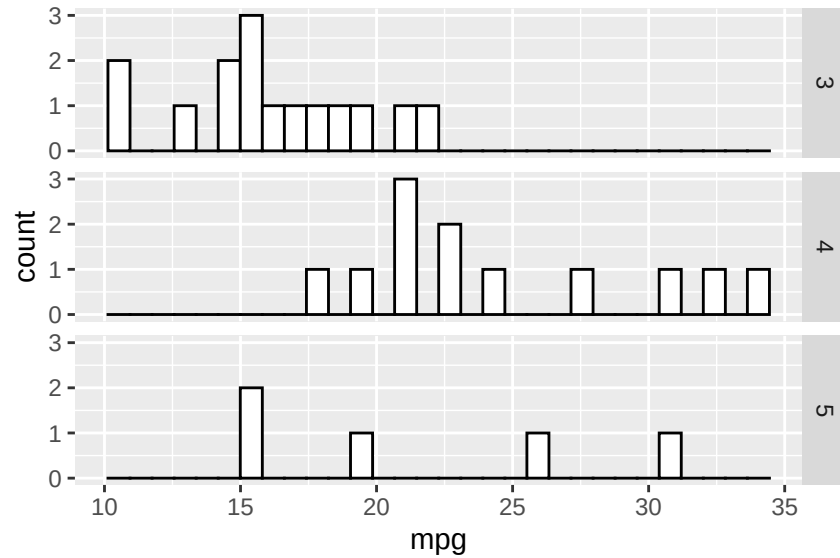


```
# Change colors
ggplot(my_data, aes(x=mpg)) +
  geom_histogram(color="black", fill="orange")
```



The `facet_grid(...)` and `facet_wrap(...)` functions of the ggplot package provide an elegant way to plot several subsets of a variable at the same time. Below three plot of the miles-per-gallon variables, split by the number of gears.

```
ggplot(my_data, aes(x=mpg))+
  geom_histogram(color="black", fill="white")+
  facet_grid(gear ~ .)
```



Exercise 2

Make a plot that separates horsepower ('hp') into vertical panels by manual/automatic transmission ('am' variable). The bin should have a width of 40.

Bar plots

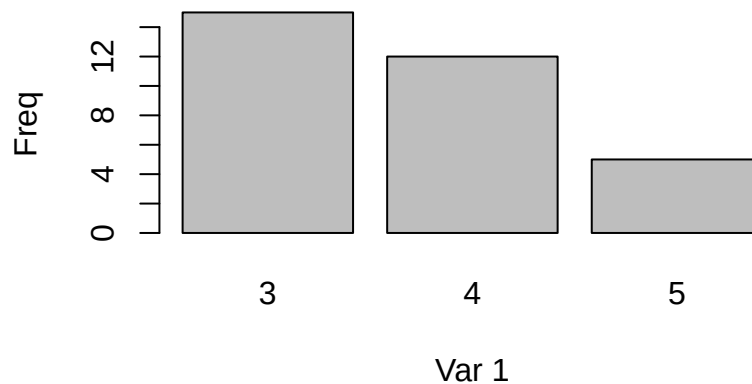
The `barplot(...)` command allows to draw simple bar plots. However, first we need to manually calculate the frequency table using the `table()` command.

For example of the data shown in the table below:

Var1	Freq
3	15
4	12
5	5

Can be plotted with:

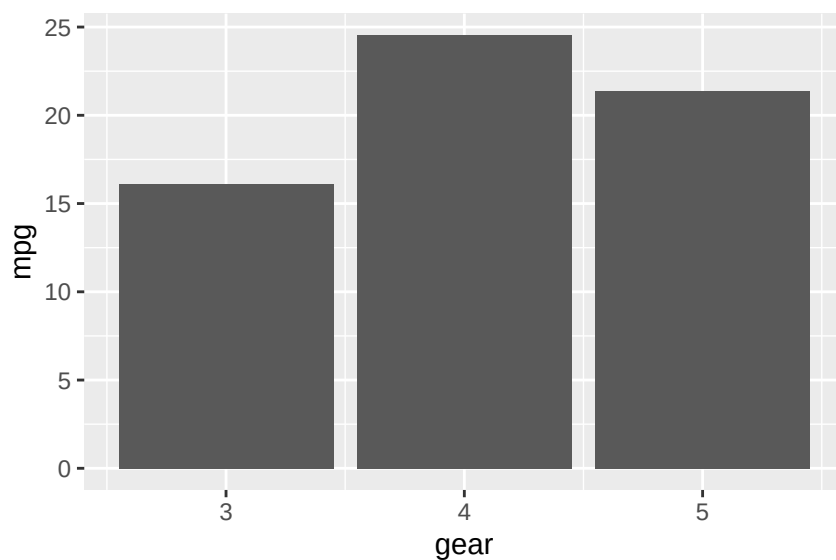
```
counts <- table(my_data$gear)
barplot(counts, xlab="Var 1", ylab = "Freq")
```



Bar plots with ggplot

However, most of the time we are interested in plotting means rather than counts. The 'ggplot' package allows to create the bar plot without the need to first calculate the summary statistics:

```
library(ggplot2)
ggplot(my_data, aes(x=gear, y=mpg)) + geom_bar(stat = "summary", fun = "mean")
```



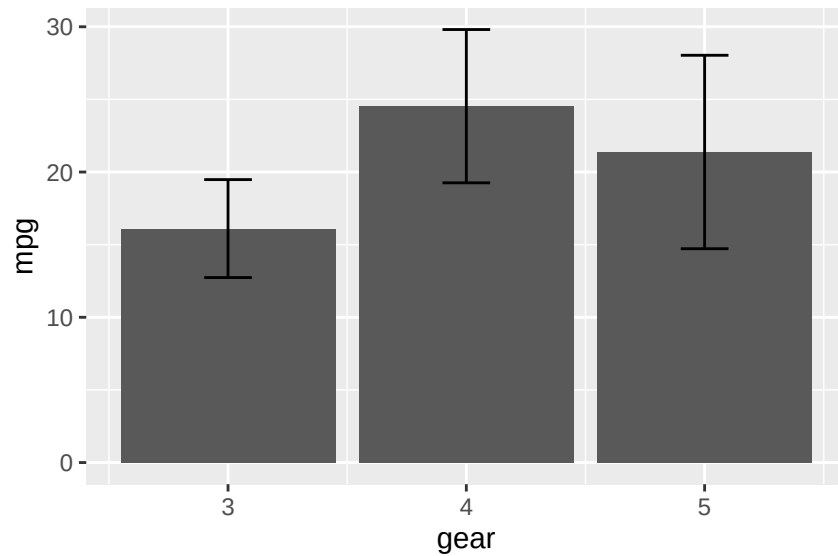
If we want to add error bars, we unfortunately first have to compute the summary statistics ourselves before we can pass the data to the ggplot command

```
library(plyr)

## Warning: package 'plyr' was built under R version 4.2.2
df <- ddply(my_data, c("gear"), summarize, Mean = mean(mpg), SD = sd(mpg))

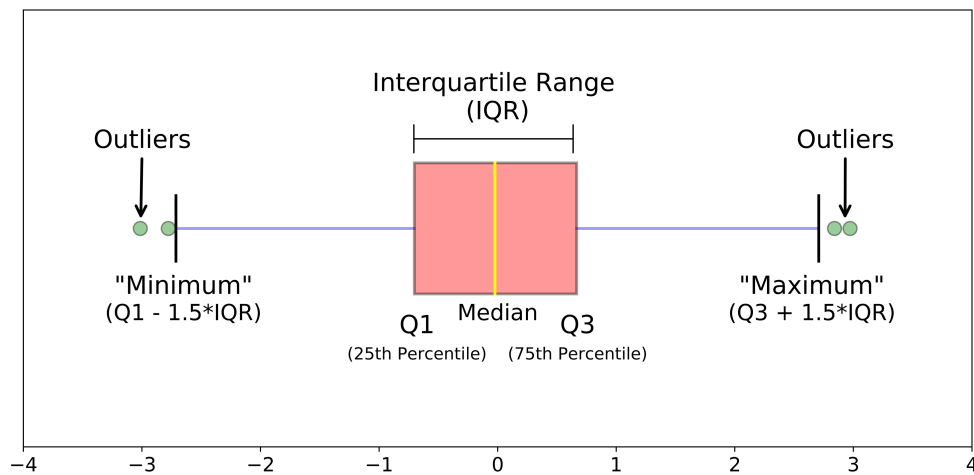
ggplot(df, aes(x = gear, y = Mean)) +
```

```
geom_bar(stat = "identity") +
  ylab("mpg") +
  geom_errorbar(aes(ymin = Mean - SD, ymax = Mean + SD), width = 0.2)
```



Box plot

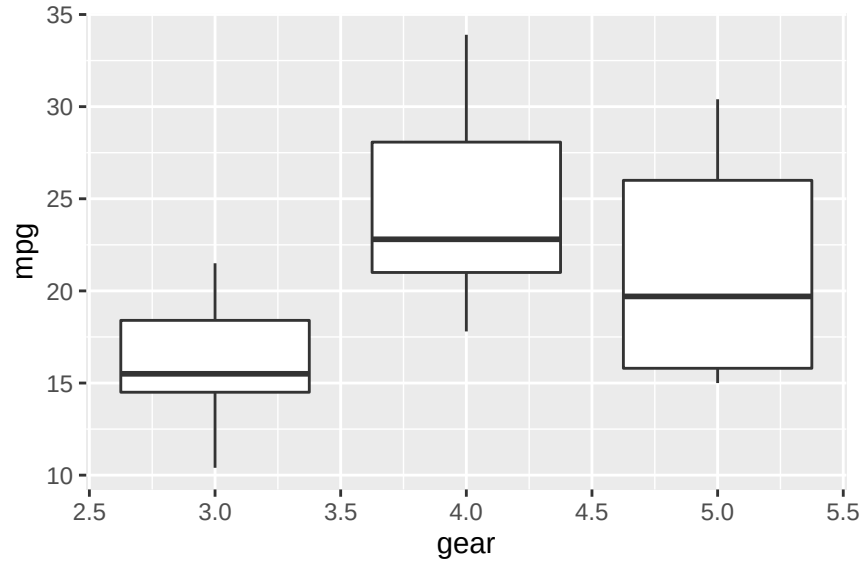
The detour we have to take to first calculate the summary statistics, is one of the reason box plot are popular. Box plots contain more information than a bar plot with errorbars. The image below shows how to read



a box plot. (source: <https://towardsdatascience.com/understanding-boxplots-5e2df7bcbd51>) Note that box plot are common in the exploratory phase of data analysis, but are not commonly used in publications.

With ggplot, we can do the box plot in a single step:

```
ggplot(my_data, aes(x=gear, y=mpg)) + geom_boxplot(aes(group=gear))
```



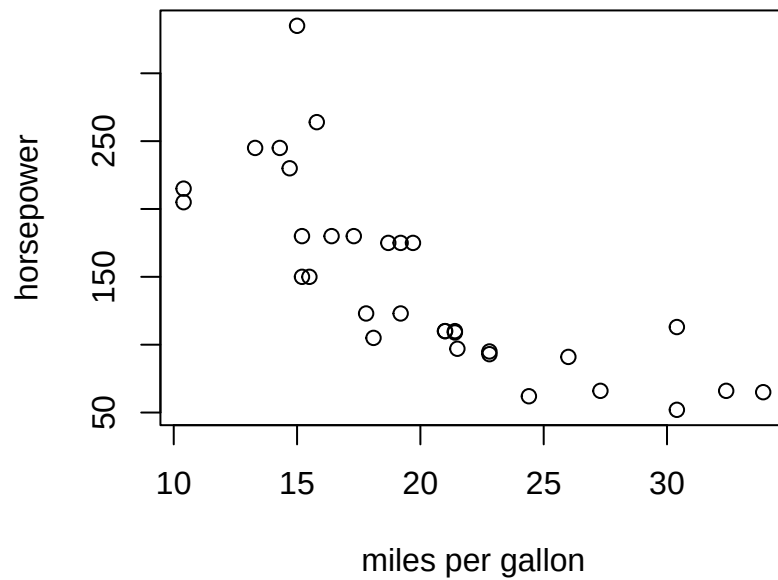
Exercise 3

Create both, a bar plot and a box plot, that compare the horsepower ('hp') of vehicles with different numbers of cylinders ('cyl'). Add a meaningful label to the x- and y-axis.

Scatter plot

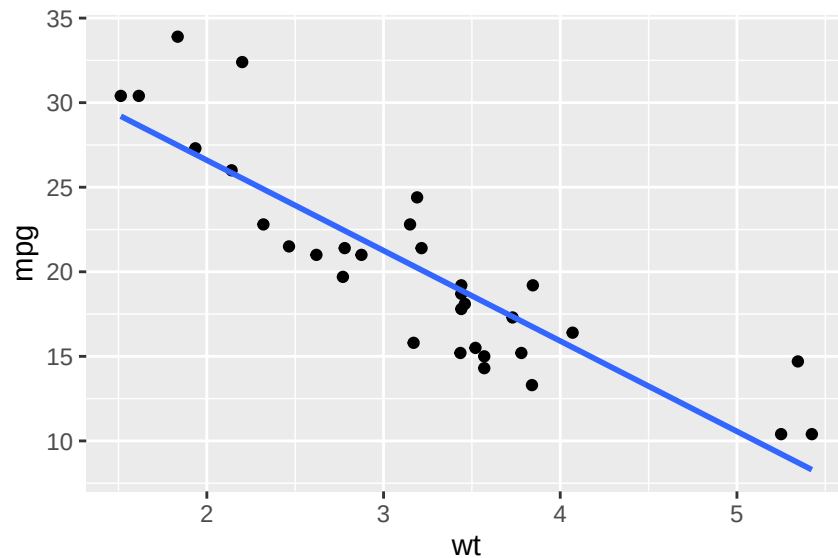
So far we have only looked at individual variables. If we want to visualise the relationship between two variables, we can do this using a scatter plot. The `plot(...)` function does this if we provide it with two variables:

```
plot(my_data$mpg, my_data$hp, xlab="miles per gallon", ylab="horsepower")
```

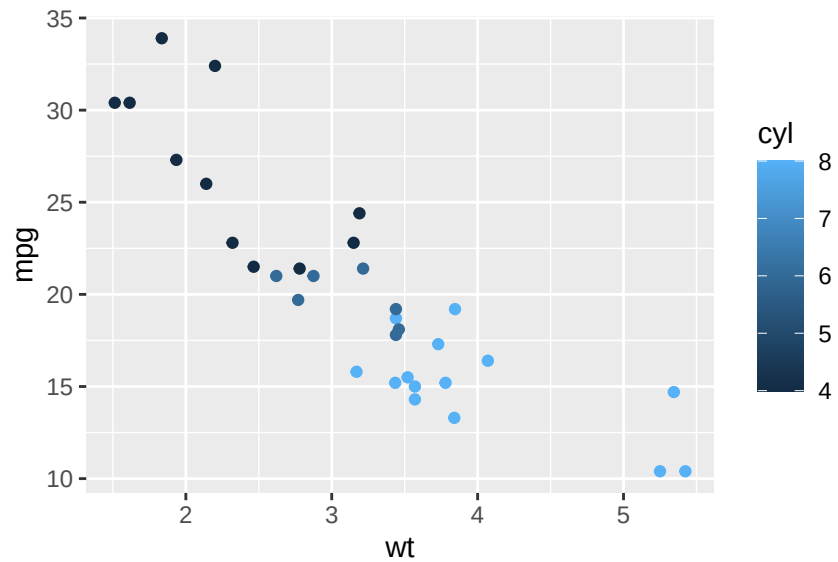
Using the ggplot package we can easily add a regression line to the scatter plot:

```
ggplot(my_data, aes(x=wt, y=mpg)) +  
  geom_point()+  
  geom_smooth(method=lm, se=FALSE)
```



Going one step further we can colour the dots according to the number of cylinders:

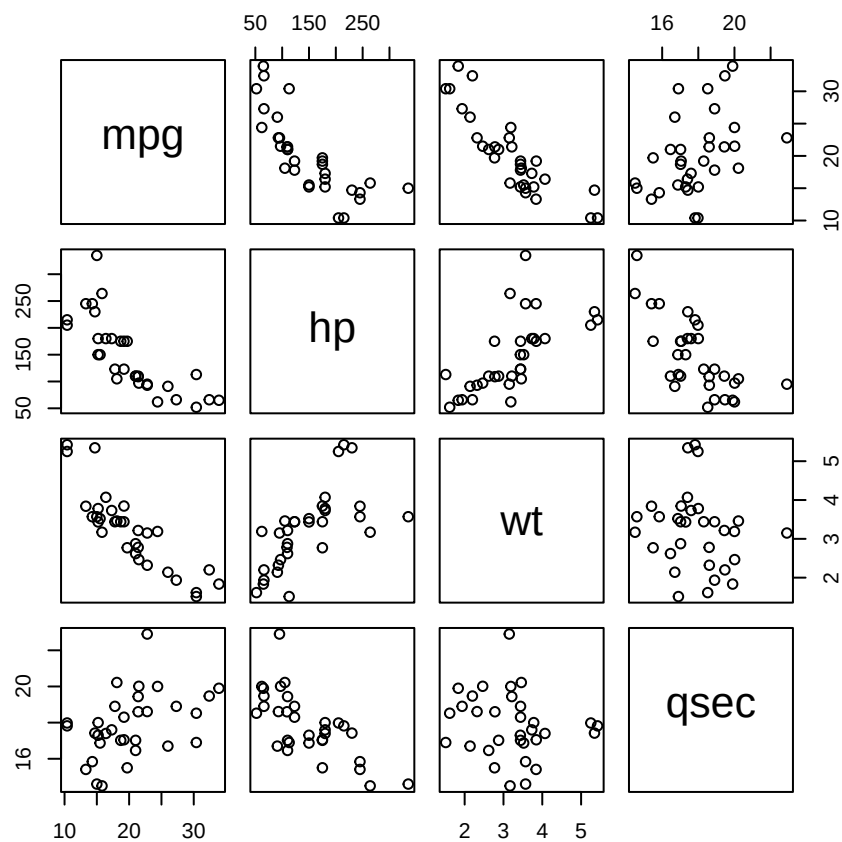
```
# Change point shapes and colors  
ggplot(my_data, aes(x=wt, y=mpg, color=cyl)) +  
  geom_point()
```



Scatter plot matrix

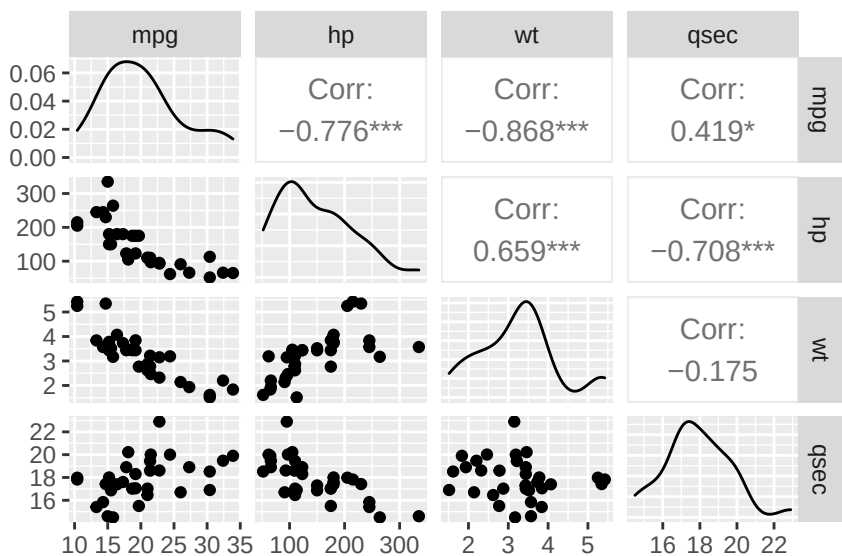
When exploring the data, we often want to look at the pair-wise relationship between variables. For example, we might be interested how miles per gallons is related to horsepower, weight, and quarter mile time. Creating a matrix of scatter plots is provided through the `pairs(...)` function:

```
pairs(~mpg+hp+wt+qsec, data=my_data)
```



A more powerful way of doing scatter plot matrices is provided by the ‘GGally’ package. If needed, install it with `install.packages("GGally")`.

```
library("GGally")
ggpairs(my_data, columns = c("mpg", "hp", "wt", "qsec"))
```



Exercise 4

Create a scatter plot of horsepower (hp) vs miles per gallon (mpg) where the dots are coloured according to the number of gears.